

Supporting Information: Anatomical prediction of sacroiliac motion and its load-dependent limits

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1 Supplementary methods and reproducibility

All supplementary statistics use the unloaded-subject reference $\mathbf{X} + \phi_i$ and loaded coordinates $\mathbf{X} + \phi_i + \mathbf{u}_{il}$, as specified in the mathematical appendix of the main article. The three simulation variants share subject indices; resampling preserves their pairing. G denotes geometry-preserved / standardized-density (individual geometry including scale, with the spatially varying population-median density field); D denotes density-preserved / template-geometry. Full denotes individual geometry and density. The standardized density field is spatially varying, not homogeneous bone.

Component differences compare bilateral absolute magnitudes, not signed motion vectors. Their bootstrap intervals describe medians; primary tests are exact binomial sign tests. Signed-rank results are retained only as machine-readable sensitivity diagnostics. The six fully adjusted sex comparisons use HC3 covariance and a single six-test FDR family. Variance ratios are non-additive and are not causal contributions.

The exploratory conditional volume-association model is

$$\begin{aligned} \log Y = & \beta_0 + \beta_v \log V + \beta_s \text{Female} + \beta_{vs} \log V \text{Female} \\ & + \beta_a \text{Age}_z + \beta_{AP} \text{AP}_z + \beta_b \text{Biischadic}_z + \beta_p \text{Subpubic}_z + \epsilon. \end{aligned} \quad (1)$$

Male and female slopes are β_v and $\beta_v + \beta_{vs}$. Conditional volume-slope tests are corrected jointly across 20 sex-specific slopes; the ten interaction tests form a separate family. Table S4 is an exact reparameterization with $\log s = (\log V - \log V_{\text{template}})/3$, so log-scale slopes equal three times the log-volume slopes. It is not an independent validation or sensitivity analysis.

The central prediction comparison uses fixed retrospective predictor sets and repeated folds that hold out subjects. Standardization, OLS coefficients and arithmetic-scale smearing are estimated only on training subjects. Gains and their intervals use paired held-out losses, resampling subjects rather than treating validation repeats as independent. The original selection of the triad is not nested in validation.

The controlled load-mixture experiment combines full FE displacement fields at fixed subject stiffness, retaining common pretension because the weights sum to one. For each side, a vector endpoint combination obeys

$$\|(1-w)\mathbf{d}_1 + w\mathbf{d}_2\|^2 = (1-w)^2\|\mathbf{d}_1\|^2 + w^2\|\mathbf{d}_2\|^2 + 2w(1-w)\mathbf{d}_1 \cdot \mathbf{d}_2. \quad (2)$$

Scalar endpoint magnitudes cannot recover the final cross-term. Motion is re-extracted from mixed fields; vector-combination error is measured because the affine/rigid fitting operations do not themselves superpose exactly. Both endpoint estimates use individual FE information and are distinct from held-out anatomical prediction.

The analysis comprises motion extraction, extraction sensitivity checks, held-out prediction, the controlled load-mixture experiment, supporting statistical comparisons, and generation of tables and figures. Motion is measured relative to each subject's unloaded configuration; resampling retains subject pairing across loads and model variants.

Table S1: Directional re-routing metrics

Load	Component	Median difference [95% CI]	Raw p	BH q
SP1leg	AP (mm)	0.024 [0.014, 0.039]	< 0.001	< 0.001
SP1leg	ML (mm)	0.144 [0.124, 0.160]	< 0.001	< 0.001
SP1leg	CC (mm)	0.439 [0.416, 0.471]	< 0.001	< 0.001
LAB1	AP (mm)	-0.303 [-0.322, -0.270]	< 0.001	< 0.001
LAB1	ML (mm)	-0.007 [-0.018, 0.002]	0.254	0.254
LAB1	CC (mm)	-0.854 [-0.886, -0.817]	< 0.001	< 0.001
LAB2	AP (mm)	-0.181 [-0.207, -0.157]	< 0.001	< 0.001
LAB2	ML (mm)	0.052 [0.047, 0.061]	< 0.001	< 0.001
LAB2	CC (mm)	-0.877 [-0.900, -0.841]	< 0.001	< 0.001
LAB3	AP (mm)	0.032 [0.018, 0.043]	< 0.001	< 0.001
LAB3	ML (mm)	0.011 [0.003, 0.017]	0.007	0.007
LAB3	CC (mm)	-0.556 [-0.586, -0.537]	< 0.001	< 0.001

Table S2: Variance ratios and paired agreement

Load	Endpoint	G/full % [95% CI]	D/full % [95% CI]
SP2leg	Rotation (deg)	103.89 [100.48, 107.29]	0.823 [0.589, 1.145]
SP2leg	Translation (mm)	100.37 [98.00, 102.93]	0.984 [0.745, 1.297]
SP2leg	AP (mm)	99.94 [98.11, 102.22]	0.233 [0.163, 0.333]
SP2leg	ML (mm)	104.66 [101.49, 107.06]	0.426 [0.217, 1.096]
SP2leg	CC (mm)	99.85 [97.58, 102.21]	1.069 [0.793, 1.464]
SP1leg	Rotation (deg)	101.77 [100.44, 103.13]	0.111 [0.074, 0.169]
SP1leg	Translation (mm)	100.29 [99.16, 101.54]	0.152 [0.110, 0.209]
SP1leg	AP (mm)	100.43 [97.90, 103.31]	1.047 [0.808, 1.361]
SP1leg	ML (mm)	101.13 [99.60, 102.64]	0.014 [0.008, 0.024]
SP1leg	CC (mm)	100.43 [99.20, 101.77]	0.244 [0.181, 0.333]
LAB1	Rotation (deg)	120.61 [110.69, 130.00]	5.789 [4.458, 7.466]
LAB1	Translation (mm)	107.43 [101.54, 113.20]	2.113 [1.641, 2.698]
LAB1	AP (mm)	100.68 [95.89, 105.48]	2.704 [2.061, 3.527]
LAB1	ML (mm)	109.31 [99.08, 118.28]	1.405 [0.965, 1.981]
LAB1	CC (mm)	107.81 [100.42, 115.61]	5.737 [4.496, 7.151]
LAB2	Rotation (deg)	102.97 [100.20, 105.68]	0.145 [0.089, 0.226]
LAB2	Translation (mm)	100.16 [97.41, 102.96]	0.126 [0.098, 0.164]
LAB2	AP (mm)	101.21 [99.17, 103.15]	0.089 [0.069, 0.115]
LAB2	ML (mm)	96.43 [92.57, 100.45]	1.098 [0.742, 1.606]
LAB2	CC (mm)	104.39 [98.60, 111.08]	0.568 [0.347, 0.873]
LAB3	Rotation (deg)	102.20 [100.80, 103.58]	0.030 [0.016, 0.052]
LAB3	Translation (mm)	99.44 [98.62, 100.24]	0.051 [0.039, 0.069]
LAB3	AP (mm)	100.15 [99.12, 101.13]	0.066 [0.047, 0.093]
LAB3	ML (mm)	98.32 [95.49, 101.45]	0.031 [0.017, 0.060]
LAB3	CC (mm)	98.97 [97.73, 100.14]	0.080 [0.061, 0.103]

Table S2 continued: geometry-preserved/full agreement

Load	Endpoint	Identity R^2	RMSE	MAE
SP2leg	Rotation (deg)	0.9794	0.0318	0.0216
SP2leg	Translation (mm)	0.9849	0.0280	0.0198
SP2leg	AP (mm)	0.9926	0.0142	0.0082
SP2leg	ML (mm)	0.9956	0.0070	0.0041
SP2leg	CC (mm)	0.9861	0.0254	0.0187
SP1leg	Rotation (deg)	0.9974	0.0223	0.0150
SP1leg	Translation (mm)	0.9975	0.0185	0.0128
SP1leg	AP (mm)	0.9877	0.0135	0.0084
SP1leg	ML (mm)	0.9984	0.0060	0.0039
SP1leg	CC (mm)	0.9972	0.0193	0.0141
LAB1	Rotation (deg)	0.8777	0.0200	0.0149
LAB1	Translation (mm)	0.9457	0.0111	0.0080
LAB1	AP (mm)	0.9795	0.0040	0.0025
LAB1	ML (mm)	0.9472	0.0052	0.0035
LAB1	CC (mm)	0.9102	0.0134	0.0099
LAB2	Rotation (deg)	0.9835	0.0147	0.0093
LAB2	Translation (mm)	0.9842	0.0113	0.0070
LAB2	AP (mm)	0.9966	0.0066	0.0048
LAB2	ML (mm)	0.9547	0.0070	0.0046
LAB2	CC (mm)	0.9510	0.0129	0.0079
LAB3	Rotation (deg)	0.9973	0.0128	0.0077
LAB3	Translation (mm)	0.9987	0.0064	0.0044
LAB3	AP (mm)	0.9984	0.0070	0.0047
LAB3	ML (mm)	0.9985	0.0020	0.0013
LAB3	CC (mm)	0.9966	0.0059	0.0041

Table S3: Fully adjusted LAB sex models

Load	Outcome	Female minus male [95% CI]	Raw p	BH q
LAB1	Rotation (deg)	-0.009 [-0.026, 0.008]	0.283	0.566
LAB1	Translation (mm)	0.013 [-0.005, 0.031]	0.166	0.566
LAB2	Rotation (deg)	-0.017 [-0.070, 0.037]	0.543	0.736
LAB2	Translation (mm)	-0.031 [-0.078, 0.016]	0.192	0.566
LAB3	Rotation (deg)	-0.021 [-0.105, 0.062]	0.613	0.736
LAB3	Translation (mm)	0.002 [-0.060, 0.064]	0.950	0.950

Table S4: Exact scaling reparameterization using log(scale)

Load	Outcome	Male scale slope [95% CI]	Female scale slope [95% CI]
SP2leg	Rotation (deg)	-2.038 [-2.670, -1.406]	-1.816 [-2.293, -1.339]
SP2leg	Translation (mm)	-1.097 [-1.592, -0.601]	-1.205 [-1.566, -0.845]
SP1leg	Rotation (deg)	-1.866 [-2.685, -1.047]	-1.649 [-2.281, -1.018]
SP1leg	Translation (mm)	-1.379 [-2.132, -0.625]	-1.191 [-1.774, -0.609]
LAB1	Rotation (deg)	-2.747 [-3.432, -2.062]	-3.186 [-3.698, -2.674]
LAB1	Translation (mm)	-1.703 [-2.776, -0.629]	-3.010 [-3.811, -2.210]
LAB2	Rotation (deg)	-1.622 [-2.142, -1.101]	-2.021 [-2.412, -1.631]
LAB2	Translation (mm)	-2.453 [-3.538, -1.368]	-2.840 [-3.861, -1.818]
LAB3	Rotation (deg)	-2.549 [-3.144, -1.954]	-3.136 [-3.720, -2.552]
LAB3	Translation (mm)	-2.100 [-2.766, -1.434]	-2.711 [-3.477, -1.946]

Table S5: Pooled models with subject-clustered covariance

Coefficients with interaction terms are conditional: the volume main effect refers to male SP2leg. Outcome-SD coefficients divide by the pooled outcome SD; binary predictors retain their coding. Log-volume and the other continuous covariates are standardized. No robust-Wald statistic is interpreted as partial variance explained.

Outcome	Predictor	β [95% CI]	Outcome-SD β	q_{FDR}
Rotation magnitude ($ \theta $, deg)	SP1leg (vs SP2leg)	1.087 [1.048, 1.126]	1.667	< 0.001
	LAB1 (vs SP2leg)	-0.746 [-0.764, -0.728]	-1.145	< 0.001
	LAB2 (vs SP2leg)	-0.241 [-0.260, -0.223]	-0.370	< 0.001
	LAB3 (vs SP2leg)	-0.104 [-0.119, -0.089]	-0.160	< 0.001
	Female sex	0.011 [-0.046, 0.068]	0.017	0.705
	Age (z)	0.018 [0.004, 0.032]	0.027	0.017
	Log pelvic volume (z)	-0.118 [-0.142, -0.094]	-0.181	< 0.001
	SP1leg $\times$ Log volume (z)	-0.115 [-0.163, -0.067]	-0.176	< 0.001
	LAB1 $\times$ Log volume (z)	0.109 [0.090, 0.128]	0.167	< 0.001
	LAB2 $\times$ Log volume (z)	0.086 [0.067, 0.104]	0.131	< 0.001
	LAB3 $\times$ Log volume (z)	-0.015 [-0.031, 0.000]	-0.024	0.069
	Inlet AP diameter (z)	-0.009 [-0.024, 0.007]	-0.013	0.286
	Biischadic width (z)	0.048 [0.031, 0.065]	0.074	< 0.001
	Subpubic angle (z)	0.016 [-0.010, 0.043]	0.025	0.254
	Female $\times$ Log volume (z)	-0.036 [-0.068, -0.005]	-0.056	0.033
Translation magnitude ($ \mathbf{d} $, mm)	SP1leg (vs SP2leg)	0.605 [0.579, 0.630]	1.014	< 0.001
	LAB1 (vs SP2leg)	-0.907 [-0.927, -0.887]	-1.520	< 0.001
	LAB2 (vs SP2leg)	-0.792 [-0.812, -0.772]	-1.328	< 0.001
	LAB3 (vs SP2leg)	-0.478 [-0.494, -0.461]	-0.801	< 0.001
	Female sex	0.046 [0.003, 0.089]	0.077	0.045
	Age (z)	0.013 [0.002, 0.024]	0.021	0.034
	Log pelvic volume (z)	-0.103 [-0.124, -0.083]	-0.173	< 0.001
	SP1leg $\times$ Log volume (z)	-0.066 [-0.096, -0.036]	-0.111	< 0.001
	LAB1 $\times$ Log volume (z)	0.118 [0.099, 0.136]	0.197	< 0.001
	LAB2 $\times$ Log volume (z)	0.109 [0.092, 0.126]	0.183	< 0.001
	LAB3 $\times$ Log volume (z)	0.041 [0.024, 0.057]	0.068	< 0.001
	Inlet AP diameter (z)	-0.004 [-0.016, 0.008]	-0.007	0.507
	Biischadic width (z)	0.067 [0.052, 0.083]	0.113	< 0.001
	Subpubic angle (z)	0.012 [-0.008, 0.033]	0.021	0.245
	Female $\times$ Log volume (z)	-0.022 [-0.045, 0.001]	-0.036	0.077

Table S6: Extraction sensitivity

Absolute differences between rigid-only and affine sacral correction in six subjects at cohort row indices 0, 55, 110, 165, 220 and 277. Each contributes five loads. These are subset maxima, not cohort-wide bounds.

Load	Number	Max $ \Delta \text{ rotation} $ (deg)	Max $ \Delta \text{ translation} $ (mm)
SP2leg	6	0.0225	0.0049
SP1leg	6	0.0047	0.0055
LAB1	6	0.0013	0.0007
LAB2	6	0.0030	0.0006
LAB3	6	0.0229	0.0016

Age functional-form sensitivity

A squared standardized-age term was added to each nested sex model (M0–M2) and each load-specific conditional log-volume model, retaining all other terms and HC3 covariance. This diagnostic used the same 278 subjects and did not replace the primary linear-age estimates. Adjustment for pelvic volume and the selected morphometric triad continued to attenuate translation contrasts: the maximum change in an M2 translation coefficient was 0.000334 mm, and all six M2 intervals included zero. All 20 conditional volume slopes remained negative with BH-adjusted $q \leq 0.002$; the largest absolute slope change was 0.0196. Multiplicity correction used the same six-coefficient and 20-slope families as the primary analyses. These checks address age functional form, not causal mediation or equivalence.

Table S7: Sex contrasts with quadratic age adjustment

Coefficients are female-minus-male differences in the outcome units. The M0 and M2 quadratic-age columns show the selected morphometric adjustment; the linear-age M2 column reproduces the primary model for comparison.

Load	Outcome	M0 (quadratic age)	M2 (linear age)	M2 (quadratic age) [95% CI]	BH q
LAB1	Rotation (deg)	+0.0735	-0.0093	-0.0092 [-0.0261, +0.0078]	0.582
LAB1	Translation (mm)	+0.0548	+0.0130	+0.0130 [-0.0055, +0.0315]	0.576
LAB2	Rotation (deg)	+0.0933	-0.0166	-0.0168 [-0.0704, +0.0368]	0.746
LAB2	Translation (mm)	+0.0714	-0.0312	-0.0316 [-0.0790, +0.0159]	0.576
LAB3	Rotation (deg)	+0.3536	-0.0214	-0.0208 [-0.1034, +0.0618]	0.746
LAB3	Translation (mm)	+0.2581	+0.0020	+0.0020 [-0.0601, +0.0641]	0.949

Table S8: Conditional volume slopes with quadratic age adjustment

Slopes are dimensionless coefficients for log motion versus log total mapped bony pelvic volume. Sex-specific slopes include the sex–volume interaction and are conditional on the selected morphometric triad.

Load	Outcome	Sex	Linear-age slope	Quadratic-age slope [95% CI]	BH q
SP2leg	Rotation	Male	-0.679	-0.677 [-0.888, -0.466]	< 0.001
SP2leg	Rotation	Female	-0.605	-0.597 [-0.755, -0.440]	< 0.001
SP2leg	Translation	Male	-0.366	-0.363 [-0.526, -0.200]	< 0.001
SP2leg	Translation	Female	-0.402	-0.392 [-0.512, -0.273]	< 0.001
SP1leg	Rotation	Male	-0.622	-0.617 [-0.888, -0.346]	< 0.001
SP1leg	Rotation	Female	-0.550	-0.530 [-0.741, -0.319]	< 0.001
SP1leg	Translation	Male	-0.460	-0.455 [-0.705, -0.205]	< 0.001
SP1leg	Translation	Female	-0.397	-0.381 [-0.576, -0.186]	< 0.001
LAB1	Rotation	Male	-0.916	-0.916 [-1.145, -0.686]	< 0.001
LAB1	Rotation	Female	-1.062	-1.063 [-1.236, -0.889]	< 0.001
LAB1	Translation	Male	-0.568	-0.569 [-0.928, -0.210]	0.002
LAB1	Translation	Female	-1.003	-1.008 [-1.280, -0.736]	< 0.001
LAB2	Rotation	Male	-0.541	-0.541 [-0.715, -0.367]	< 0.001
LAB2	Rotation	Female	-0.674	-0.677 [-0.806, -0.548]	< 0.001
LAB2	Translation	Male	-0.818	-0.822 [-1.187, -0.457]	< 0.001
LAB2	Translation	Female	-0.947	-0.963 [-1.310, -0.615]	< 0.001
LAB3	Rotation	Male	-0.850	-0.849 [-1.047, -0.651]	< 0.001
LAB3	Rotation	Female	-1.045	-1.042 [-1.235, -0.848]	< 0.001
LAB3	Translation	Male	-0.700	-0.700 [-0.923, -0.477]	< 0.001
LAB3	Translation	Female	-0.904	-0.905 [-1.161, -0.650]	< 0.001

Table S9: Exploratory conditional log-volume slopes

Male and female slopes use HC3 intervals. Slope q values use the 20-test family; interaction q values use a separate ten-test family. The models hold the selected dimensions fixed and do not describe uniformly scaled anatomies.

Load	Outcome	Male slope [95% CI]	Male q	Female slope [95% CI]	Female q	Interaction q
SP2leg	Rotation (deg)	-0.679 [-0.890, -0.469]	< 0.001	-0.605 [-0.764, -0.446]	< 0.001	0.699
SP2leg	Translation (mm)	-0.366 [-0.531, -0.200]	< 0.001	-0.402 [-0.522, -0.282]	< 0.001	0.699
SP1leg	Rotation (deg)	-0.622 [-0.895, -0.349]	< 0.001	-0.550 [-0.760, -0.339]	< 0.001	0.699
SP1leg	Translation (mm)	-0.460 [-0.711, -0.208]	< 0.001	-0.397 [-0.591, -0.203]	< 0.001	0.699
LAB1	Rotation (deg)	-0.916 [-1.144, -0.687]	< 0.001	-1.062 [-1.233, -0.891]	< 0.001	0.588
LAB1	Translation (mm)	-0.568 [-0.925, -0.210]	0.002	-1.003 [-1.270, -0.737]	< 0.001	0.343
LAB2	Rotation (deg)	-0.541 [-0.714, -0.367]	< 0.001	-0.674 [-0.804, -0.544]	< 0.001	0.494
LAB2	Translation (mm)	-0.818 [-1.179, -0.456]	< 0.001	-0.947 [-1.287, -0.606]	< 0.001	0.699
LAB3	Rotation (deg)	-0.850 [-1.048, -0.651]	< 0.001	-1.045 [-1.240, -0.851]	< 0.001	0.494
LAB3	Translation (mm)	-0.700 [-0.922, -0.478]	< 0.001	-0.904 [-1.159, -0.649]	< 0.001	0.494

Table S10: Paired held-out prediction gains and benchmarks

Metrics use ten repeats of ten-fold subject-held-out cross-validation, with the same folds for every load, outcome and predictor set. Size uses age and log volume; triad adds AP diameter, interspinous width and subpubic angle. Triad+sex adds female sex and its log-volume interaction. The eight-measure model uses age, log volume and all eight morphometric measures. Positive ΔRMSE or percentage gain favors the augmented model. RMSE units are degrees for rotation and mm for translation. Intervals are pointwise paired subject-bootstrap intervals conditional on the fitted predictions, without refitting, feature-selection uncertainty or multiplicity adjustment.

Load	Outcome	ΔRMSE size-triad [95% CI]	$R^2_{\text{triad+sex}}$	$R^2_{\text{8 measures}}$	Sex gain (%)	8-measure gain (%)
SP2leg	Rotation	0.0213 [0.0130, 0.0297]	0.611	0.606	0.0	-0.6
SP2leg	Translation	0.0543 [0.0421, 0.0663]	0.729	0.718	2.1	0.3
SP1leg	Rotation	0.0109 [0.0006, 0.0211]	0.400	0.407	-1.1	-0.6
SP1leg	Translation	0.0490 [0.0327, 0.0653]	0.517	0.514	-0.5	-0.7
LAB1	Rotation	0.0065 [0.0043, 0.0088]	0.682	0.681	-0.7	-0.9
LAB1	Translation	0.0077 [0.0048, 0.0104]	0.490	0.522	0.2	3.4
LAB2	Rotation	0.0011 [-0.0012, 0.0035]	0.412	0.407	-0.5	-0.9
LAB2	Translation	0.0066 [0.0024, 0.0108]	0.304	0.305	-0.3	-0.2
LAB3	Rotation	0.0456 [0.0331, 0.0590]	0.687	0.699	-0.6	1.4
LAB3	Translation	0.0462 [0.0341, 0.0578]	0.670	0.681	-0.8	0.9

Table S10 continued: Additional predictor gains and individual error

Sex and eight-measure gains compare each benchmark with the triad. MAE and the 95th percentile of absolute error pool held-out errors across repeats for the triad model, rather than errors of repeat-averaged predictions. No benchmark gain interval is wholly positive. Nineteen of twenty intervals include zero; the remaining interval favors the triad over the sex-augmented model for SP1leg rotation.

Load	Outcome	Sex ΔRMSE [95% CI]	8-measure ΔRMSE [95% CI]	Triad MAE	Triad P95 error
SP2leg	Rotation	-0.00004 [-0.00249, 0.00237]	-0.00086 [-0.00370, 0.00216]	0.1071	0.2550
SP2leg	Translation	0.00258 [-0.00107, 0.00616]	0.00039 [-0.00214, 0.00301]	0.0957	0.2425
SP1leg	Rotation	-0.00364 [-0.00795, -0.00002]	-0.00187 [-0.00903, 0.00521]	0.2626	0.6119
SP1leg	Translation	-0.00122 [-0.00626, 0.00342]	-0.00187 [-0.00780, 0.00376]	0.1972	0.5158
LAB1	Rotation	-0.00022 [-0.00082, 0.00028]	-0.00028 [-0.00091, 0.00031]	0.0232	0.0649
LAB1	Translation	0.00008 [-0.00072, 0.00092]	0.00116 [-0.00021, 0.00256]	0.0268	0.0678
LAB2	Rotation	-0.00042 [-0.00139, 0.00049]	-0.00078 [-0.00230, 0.00077]	0.0694	0.1676
LAB2	Translation	-0.00024 [-0.00094, 0.00039]	-0.00018 [-0.00170, 0.00128]	0.0574	0.1585
LAB3	Rotation	-0.00086 [-0.00262, 0.00098]	0.00187 [-0.00167, 0.00544]	0.1060	0.2566
LAB3	Translation	-0.00078 [-0.00267, 0.00084]	0.00093 [-0.00187, 0.00366]	0.0792	0.2029

Table S11: Density-standardization agreement and error tails

Errors are geometry-preserved minus full-model endpoints, paired within subject and load. This simplification retains complete individual modeled geometry, including size. Mean bias is signed; the 95th percentile and maximum use absolute errors. Cohort RMSE does not bound individual error.

Load	Outcome	RMSE	Mean bias	P95 abs. error	Max abs. error
SP2leg	Rotation (deg)	0.0318	0.0101	0.0580	0.2372
SP2leg	Translation (mm)	0.0280	0.0071	0.0486	0.2195
SP1leg	Rotation (deg)	0.0223	0.0085	0.0396	0.1446
SP1leg	Translation (mm)	0.0185	0.0055	0.0328	0.1157
LAB1	Rotation (deg)	0.0200	0.0042	0.0428	0.0650
LAB1	Translation (mm)	0.0111	0.0027	0.0234	0.0452
LAB2	Rotation (deg)	0.0147	0.0072	0.0315	0.1158
LAB2	Translation (mm)	0.0113	0.0030	0.0229	0.0861
LAB3	Rotation (deg)	0.0128	0.0060	0.0254	0.0767
LAB3	Translation (mm)	0.0064	0.0011	0.0124	0.0423

Table S12: Direct application-site contrasts

Contrasts are within-subject LAB2 minus LAB1 differences in bilateral absolute translation components. Pointwise percentile intervals use 3000 subject bootstrap samples. Exact two-sided sign tests are adjusted within the three-component family. These retrospective contrasts compare the sites directly, separately from their contrasts with SP2leg.

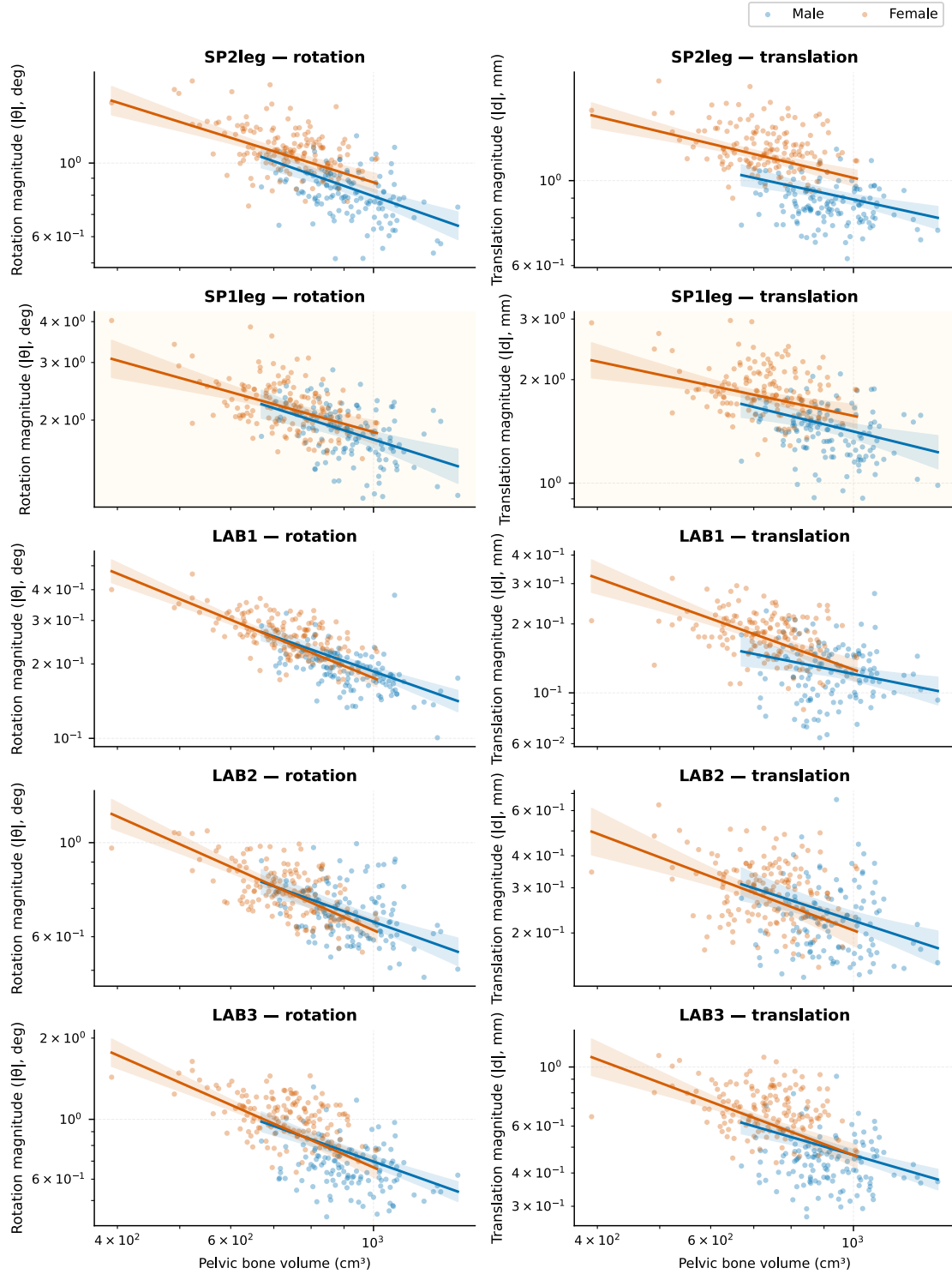
Component	LAB2-LAB1 (mm)	95% CI (mm)	<i>q</i>
ML	0.0576	[0.0536, 0.0634]	5.15e-58
AP	0.1108	[0.0879, 0.1315]	4.58e-36
CC	-0.0204	[-0.0309, -0.0125]	1.85e-06

Table S13: Effect of triad adjustment on volume slopes

Both models use log outcome, log volume, sex, sex–volume interaction and age; the adjusted model adds the triad. Intervals use HC3 covariance. The last column applies BH correction across the 20 sex-specific slopes from the model without the triad; adjusted-model q values are in Table S9. All point estimates are negative in both specifications, but precision and magnitude depend on adjustment. These conditional coefficients are distinct from held-out predictive performance.

Load	Outcome	Sex	Without triad [95% CI]	With triad [95% CI]	q without triad
SP2leg	Rotation	Male	-0.582 [-0.792, -0.372]	-0.679 [-0.890, -0.469]	1.66e-07
SP2leg	Rotation	Female	-0.535 [-0.685, -0.385]	-0.605 [-0.764, -0.446]	1.47e-11
SP2leg	Translation	Male	-0.231 [-0.389, -0.073]	-0.366 [-0.531, -0.200]	0.00502
SP2leg	Translation	Female	-0.300 [-0.437, -0.162]	-0.402 [-0.522, -0.282]	3.98e-05
SP1leg	Rotation	Male	-0.625 [-0.875, -0.376]	-0.622 [-0.895, -0.349]	2e-06
SP1leg	Rotation	Female	-0.599 [-0.780, -0.419]	-0.550 [-0.760, -0.339]	3.08e-10
SP1leg	Translation	Male	-0.345 [-0.582, -0.109]	-0.460 [-0.711, -0.208]	0.00502
SP1leg	Translation	Female	-0.324 [-0.513, -0.136]	-0.397 [-0.591, -0.203]	0.00117
LAB1	Rotation	Male	-0.680 [-0.925, -0.434]	-0.916 [-1.144, -0.687]	1.66e-07
LAB1	Rotation	Female	-0.706 [-0.849, -0.563]	-1.062 [-1.233, -0.891]	3.91e-21
LAB1	Translation	Male	-0.199 [-0.568, 0.170]	-0.568 [-0.925, -0.210]	0.291
LAB1	Translation	Female	-0.481 [-0.743, -0.219]	-1.003 [-1.270, -0.737]	0.000536
LAB2	Rotation	Male	-0.437 [-0.611, -0.264]	-0.541 [-0.714, -0.367]	2e-06
LAB2	Rotation	Female	-0.581 [-0.691, -0.471]	-0.674 [-0.804, -0.544]	1.15e-23
LAB2	Translation	Male	-0.382 [-0.743, -0.021]	-0.818 [-1.179, -0.456]	0.0424
LAB2	Translation	Female	-0.522 [-0.830, -0.214]	-0.947 [-1.287, -0.606]	0.00127
LAB3	Rotation	Male	-0.464 [-0.703, -0.224]	-0.850 [-1.048, -0.651]	0.000267
LAB3	Rotation	Female	-0.614 [-0.772, -0.456]	-1.045 [-1.240, -0.851]	1.71e-13
LAB3	Translation	Male	-0.224 [-0.470, 0.022]	-0.700 [-0.922, -0.478]	0.0786
LAB3	Translation	Female	-0.382 [-0.608, -0.155]	-0.904 [-1.159, -0.649]	0.00127

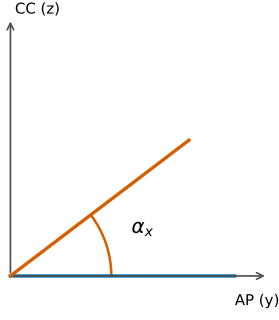
Supplementary Figure S1: Raw data and adjusted curves



Points are raw observations. Lines and bands are conditional geometric means and pointwise HC3 95% confidence intervals from the adjusted model, with age and the morphometric triad held at their cohort means. Thus the curves need not pass through the raw group means. Lines are restricted to each sex's observed volume range. The corresponding conditional slopes and intervals are reported in Supplementary Figure S3 and Table S9.

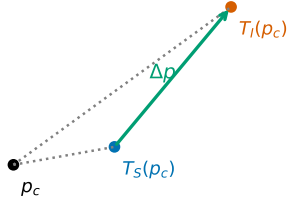
Supplementary Figure S2: Motion coordinates and geometric size

A. ML-axis rotation



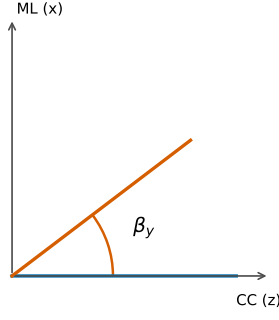
Isolated positive rotation; schematic

D. Translation at a common point



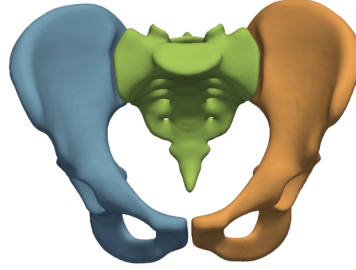
$$\mathbf{d} = \mathbf{P}^T [T_I(p_c) - T_S(p_c)]$$

B. AP-axis rotation



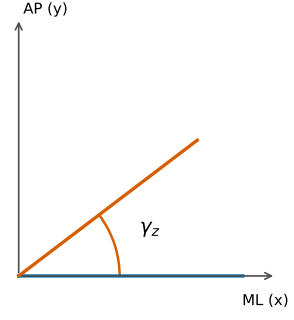
Isolated positive rotation; schematic

E. Bone geometry included



Left and right hip bones + sacrum

C. CC-axis rotation



Isolated positive rotation; schematic

F. Size and surface measures

$$V = \sum_{b=1}^3 V_b$$

$$A = \sum_{b=1}^3 A_b$$

$$s = (V/V_{\text{template}})^{1/3}$$

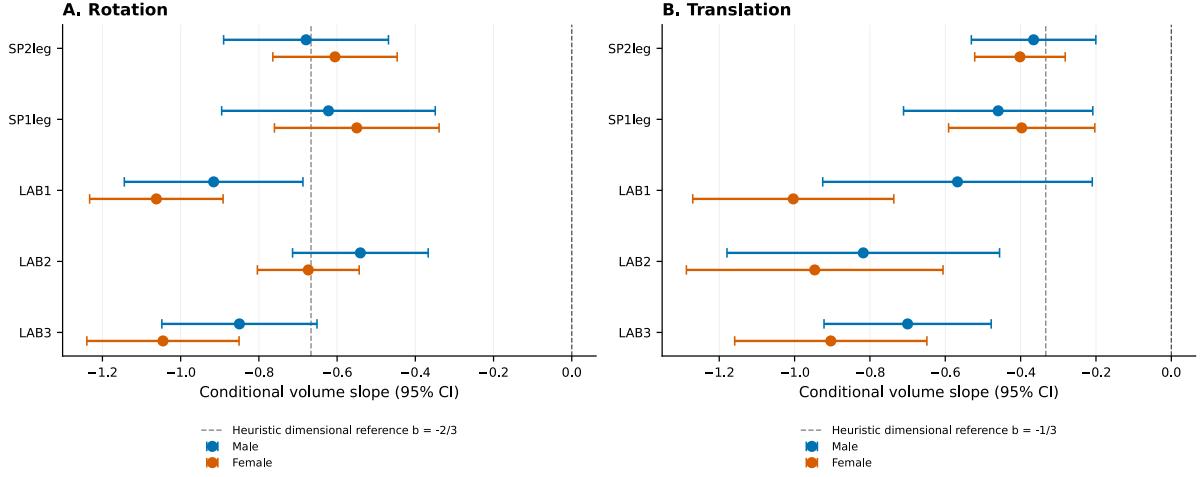
V: enclosed bone volume (cm³)

A: bone surface area (cm²)

s: dimensionless scale

(A–C) Isolated positive rotations about the ML, AP and CC axes in a right-handed reporting frame. These schematic angles are exaggerated for clarity; the actual Cardan decomposition follows $R_z(\gamma_z)R_y(\beta_y)R_x(\alpha_x)$, rather than three independent planar angle measurements. (D) Both fitted bone maps are evaluated at the same unloaded sacral ROI centroid p_c ; their difference is expressed in the pelvic frame. T_I and T_S denote the iliac and sacral rigid fits after affine correction. This is not a surface gap. (E,F) The two hip bones and sacrum define the summed enclosed bone volume and surface area; lumbar vertebrae and the space inside the pelvic canal are excluded. Scale is the cube root of the subject-to-template volume ratio. Morphometric landmark definitions are shown in main Figure 1.

Supplementary Figure S3: Conditional log-volume slopes



Male and female slopes with pointwise HC3 95% intervals from the adjusted models in Table S9. Dashed lines at $-2/3$ (rotation) and $-1/3$ (translation) are heuristic dimensional references for geometrically similar linear-elastic structures at fixed force and modulus: $u \sim F/(EL) \propto V^{-1/3}$ and $\theta \sim u/L \propto V^{-2/3}$. They are not expected conditional regression coefficients, fitted values or tested scaling laws. The regression holds other dimensions fixed, and the model's fixed ligament stiffness and nominal tissues do not enforce geometric similarity. Numerical proximity to these references therefore provides only exploratory context. Between-load slope comparisons are descriptive. The zero line indicates no conditional association; slope tests are in Table S9.